

The mathematics of removing the cut: exactness, convex cost, and where the ceiling comes from

Theory supplement to the FLEX-UF early-exit decoder. Notation follows `paper/theory.tex`; this document extends it to the regime that document's first assumption excludes.

Abstract. The tiled multi-exit decoder is analysed in the main paper under two assumptions: the compute of a frame is the mean of per-tile costs, and the distortion is the mean of per-tile distortions. The first is exact for a tiled decoder and false for a decoder that does not tile. We replace tiling with a depth field and a max-plus dilation, and prove four things. (i) The decode is *exact*: every position receives what a full-frame decode to its own depth would have produced, with no value read from outside the frame (Theorem 1). (ii) The distortion becomes exactly separable in the depth field (Theorem 2), which is what makes the oracle table computable from K full-frame decodes and is false under tiling. (iii) The compute stops being additive and becomes a convex function of the depth field, whose linearisation is precisely the additive proxy the allocator uses; the gap is the band, and it bounds how far the practical allocator can be from optimal (Theorem 4). (iv) The reachable saving is capped by a floor that the split depth inflates, and removing the split moves the ceiling from 40.1% to 69.9% (Proposition 8). A fifth result explains an empirical finding: when the exit-error curve is rank-1, the oracle allocation is a threshold rule on a single per-cell statistic, so a router needs the right *ranking* and nothing else (Theorem 6).

1 Setup

Let Ω be the feature grid of one frame and let the trunk be a composition of L blocks $G_1, \dots, G_L$, each a 3×3 operation, so that block b at position x reads only positions within Chebyshev distance 1 of x in the state after block $b-1$. Write F_b for the full-frame state after b blocks, computed everywhere. In the measured system $L = 12$, exits are taken every $\beta = 2$ blocks, and $K = 6$.

A **depth field** is a map $d: \Omega \rightarrow \{0, \dots, L\}$ giving the number of blocks each position is allocated. Tiling is the special case in which d is constant on each tile. Nothing below assumes that.

2 The dilation

Define

$$D(x) = \max_{y \in \Omega} [d(y) - \text{dist}(x, y)],$$

with dist the Chebyshev distance on the grid. In max-plus algebra this is the grey-scale morphological dilation of d by a cone of unit slope; the identification is standard and gives the next lemma for free, but the proof is two lines so it is given.

Lemma 1 (least 1-Lipschitz majorant). *D is the pointwise smallest function that dominates d and is 1-Lipschitz. That is: (i) $D \geq d$; (ii) $|D(x) - D(x')| \leq \text{dist}(x, x')$; and (iii) if $f \geq d$ is 1-Lipschitz then $f \geq D$.*

Proof. (i) take $y = x$. (ii) for any y , $\text{dist}(x, y) \leq \text{dist}(x', y) + \text{dist}(x, x')$, so $d(y) - \text{dist}(x, y) \geq d(y) - \text{dist}(x', y) - \text{dist}(x, x')$; take the max over y and repeat with the roles exchanged. (iii) 1-Lipschitz gives $f(x) \geq f(y) - \text{dist}(x, y) \geq d(y) - \text{dist}(x, y)$ for every y . ■

Property (ii) is the one that does the work below: the dilated field cannot fall by more than one block per step, which is exactly the rate at which a 3×3 receptive field grows.

3 Exactness

Let the decoder compute block b on the set $S_b = \{x : D(x) \geq b\}$ and nowhere else, leaving positions outside S_b holding their value from block $b-1$. Write $\Phi_b(x)$ for the value this procedure holds at x after block b .

Theorem 1 (exact per-position decoding). *For every b and every x with $D(x) \geq b$, $\Phi_b(x) = F_b(x)$. In particular each position's read-out at its own depth, $\Phi_{d(x)}(x)$, equals $F_{d(x)}(x)$: what a full-frame decode to depth $d(x)$ would have produced there.*

Proof. Induction on b . For $b = 0$ the state is the upsampled latent, computed everywhere, so the claim holds. Assume it for $b-1$ and take x with $D(x) \geq b$. Block b at x reads only positions y with $\text{dist}(x,y) \leq 1$. By Lemma 1(ii), $D(y) \geq D(x)-1 \geq b-1$, so every such y lies in S_{b-1} and by the induction hypothesis holds $F_{b-1}(y)$. The block therefore receives exactly the inputs the full-frame computation would give it, and being a deterministic function of them, returns $F_b(x)$. The final claim follows from $d \leq D$ (Lemma 1(i)). ■

Corollary 2 (no invented values). *The procedure never reads a value outside Ω , and in particular never reads a padded one. A tiled decoder reads one padded value per border position per block.*

Corollary 2 is the formal content of the phrase “there is no seam”. It is not that the seam is small; it is that the quantity a seam measures does not exist in this procedure, at any granularity of d .

4 Distortion becomes separable

Under tiling, the reconstruction on tile i depends on the depths of neighbouring tiles, because the padding a border reads is a function of where the cut was made. That is why the main paper’s Assumption 2 is stated as an assumption. Under Theorem 1 it becomes a consequence.

Theorem 2 (separability). *Let the read-out at x be $H(A_{d(x)} \Phi_{d(x)}(x))$ with A_k the exit- k adapter and H the head applied pointwise. Then the frame distortion is separable in the depth field:*

$$D(d) = \sum_{x \in \Omega} m(x, d(x)), \quad m(x, k) = \|H(A_k F_k(x)) - t(x)\|^2,$$

where t is the target. In particular $m(\cdot, k)$ is obtained from K full-frame decodes, one per exit, and does not depend on the allocation at all.

Proof. immediate from Theorem 1: $\Phi_{d(x)}(x) = F_{d(x)}(x)$, a quantity determined by the latent and by $d(x)$ alone. ■

The head’s receptive field. The deployed decoder does not apply the head pointwise: it stitches the adapted features and runs one head over the frame, and that head has its own 3×3 . Separability is therefore exact for the trunk output and holds for the pixel output up to a coupling supported on the one-cell neighbourhood of a depth boundary. That residual is not assumed away; it is measured. On all 53 test frames at 64 px cells it is below 0.0002 dB, and its sign is favourable — the deployed decoder is very slightly better than the separable idealisation, not worse.

Theorem 2 is what makes the rest of this project computable. Every oracle number quoted anywhere comes from tabulating $m(x, k)$ once and minimising over allocations; under tiling that table does not exist, because m would depend on the neighbours’ depths as well.

5 Cost stops being additive and becomes convex

The number of block-evaluations the procedure performs is

$$C_{\text{true}}(d) = \sum_{b=1}^L |S_b| = \sum_{x \in \Omega} D(x),$$

against the additive proxy the allocator actually optimises,

$$C_{\text{sep}}(d) = \sum_{x \in \Omega} d(x) \leq C_{\text{true}}(d),$$

with the inequality from Lemma 1(i). The difference is what this project calls the **band**. The next two results say what kind of object each side is.

Proposition 3 (level sets of the dilation). *For every integer b ,*

$$S_b = \{x : D(x) \geq b\} = \bigcup_{c \geq b} (L_c \oplus B_{c-b}), \quad L_c = \{y : d(y) \geq c\},$$

where $\oplus$ is dilation by the ball B_r of radius r .

Proof. $D(x) \geq b$ holds iff there is a y with $d(y) \geq b + \text{dist}(x, y)$. Put $c = d(y)$; then $c \geq b$ and $\text{dist}(x, y) \leq c - b$, i.e. $x \in L_c \oplus B_{c-b}$. Conversely such an x admits a witness y with $d(y) - \text{dist}(x, y) \geq c - (c - b) = b$. ■

Proposition 3 is the formal version of “the band’s width is the depth *difference* between neighbours”: a region allocated depth c forces block b to run only within $c - b$ of it, so a map whose depth varies slowly pays almost nothing. Together with a perimeter bound $|A \oplus B_r| \leq |A| + rP(A) + O(r^2)$ it gives the band a bound in terms of the perimeters of the depth level sets, which is why the measured band grows roughly with the inverse of the cell side: 0.12 points of saving at 256 px against 0.79 at 32 px, at the lowest rate.

Theorem 4 (convexity, and an *a-posteriori* optimality bound). *Regard d as a vector in $\mathbb{R}^{|\Omega|}$. Then C_{true} is convex, C_{sep} is linear, and C_{sep} is the largest linear minorant of C_{true} that is exact on constant fields. Moreover, if $d \blacksquare$ minimises the separable Lagrangian $J_{\text{sep}}(d) = D(d) + \lambda C_{\text{sep}}(d)$, then*

$$J_{\text{true}}(\hat{d}) - \min_d J_{\text{true}}(d) \leq \lambda \cdot \text{band}(\hat{d}).$$

Proof. $D(\cdot)$ is a pointwise maximum of affine functions of d , hence convex, and a sum of convex functions is convex; C_{sep} is a sum of coordinates, hence linear; on a constant field $D \equiv d$ so the two agree. For the bound, write $J_{\text{true}} = J_{\text{sep}} + \lambda \text{band}$. Then $\min_d J_{\text{true}} \geq \min_d J_{\text{sep}} = J_{\text{sep}}(d \blacksquare)$, and $J_{\text{true}}(d \blacksquare) = J_{\text{sep}}(d \blacksquare) + \lambda \text{band}(d \blacksquare)$. Subtract. ■

The bound is computable after the fact, because the band of the chosen map is measured by dilating it. It also predicts a negative result this project obtained empirically: a smoothness-regularised allocation, which is an attempt to spend less band, cannot gain more than λband and was measured to gain less than nothing — it halved the band and lost slightly more in distortion.

6 The allocation

With Theorem 2 the distortion is separable and with C_{sep} the cost is too, so the Lagrangian decouples and the per-cell argmin is exactly optimal for it. This is Everett’s theorem specialised to our setting, and it is the same argument the main paper’s Theorem 2 makes for tiles; what is new is that here it is exact rather than an assumption.

Proposition 5 (decoupling). *For every $\lambda \geq 0$,*

$$\min_d [D(d) + \lambda C_{\text{sep}}(d)] = \sum_{x \in \Omega} \min_{k \in \mathcal{K}} [m(x, k) + \lambda c_k],$$

attained by $d \blacksquare(x) \in \text{argmin}_k [m(x, k) + \lambda c_k]$. Sweeping λ traces the lower convex envelope of the achievable (cost, distortion) set, and only that envelope; points strictly inside a facet are reached by mixing two λ values, not by any single one.

A caution the implementation had to learn. The feasible set is $K = \{j, \dots, K-1\}$ and the deployed `forward()` clamps a map into it. Taking the argmin over all K exits and clamping afterwards is *not* the same optimisation: it prices the shallow options at costs c_0, c_1 that no decoder charges, and biases every allocation shallow. The two coincide only when the cost vector is itself clamped, which the shipped `exit_costs` does and a hand-written block-count does not.

6.1 Why the ranking is all a router needs

The empirical finding that motivates this subsection: on the test set the log of the exit-error curve is rank-1 to 91.7% of its variance, and a parameter-free surrogate that only ranks cells attains 83–100% of the oracle’s saving. The following explains why that is not a coincidence.

Theorem 6 (threshold structure under a rank-1 ladder). *Suppose $m(x, k) = s(x)\phi_k$ with ϕ strictly decreasing and c strictly increasing in k . Then $k^*(x) = \text{argmin}_k [s(x)\phi_k + \lambda c_k]$ is non-decreasing in $s(x)$. Consequently the oracle allocation is a threshold rule: there are cut points $\tau_j \leq \dots \leq \tau_{K-2}$ such that $k^*(x)$ is determined by which interval $s(x)$ falls in.*

Proof. for $k' > k$ the difference of the two objectives is $\Delta(s) = s(\phi_{k'} - \phi_k) + \lambda(c_{k'} - c_k)$. Since $\phi_{k'} < \phi_k$, Δ is strictly decreasing in s , so it changes sign at most once and only from + to −. This is the single-crossing condition, and the monotone comparative statics conclusion follows. ■

The practical content: under a rank-1 ladder a router does not need a calibrated estimate of the error at all. It needs the correct *order* of cells by s , plus $K-j-1$ thresholds, which the λ -sweep supplies. Any strictly monotone transform of s is as good as s . That is precisely the freedom a rate-rank surrogate exploits, and it is why replacing it with a regression that predicts s accurately buys so little: the measured gain of the best learned model over the rank-1 surrogate is 0.68 points against an oracle gap of 3.3.

Proposition 7 (misallocation costs the Lagrangian gap, not the error rate). *If a router assigns $k^\star(x)$ instead of $k^*(x)$, the excess objective is*

$$\mathcal{E} = \sum_{x \in \Omega} ([m(x, \hat{k}) + \lambda c_{\hat{k}}] - [m(x, k^*) + \lambda c_{k^*}]) \geq 0,$$

which vanishes on every cell where the two exits are tied, however many such cells are mislabelled. Agreement with the oracle is therefore not the loss a router should be measured on, and a router with lower agreement can have lower excess. Every router number in the companion report is scored by the allocation it causes, not by agreement.

7 The floor, the ceiling, and when the ceiling is out of reach

Everything above concerns which allocation is best. This section is about what the best one can possibly be worth, which is fixed by the cheapest legal allocation.

Proposition 8 (floor and ceiling). *Let the feasible exits be $\{j, \dots, K-1\}$. The cost of any allocation is at least*

$$\Phi(j) = u + (j+1)\beta p + a + h + \sigma,$$

— the upsample u , the $(j+1)\beta$ trunk blocks every position must run, the adapter a , the head h , and the seam-repair share σ —

with β blocks per exit, p the per-block share and σ the seam-repair share, and the saving is therefore at most $1 - \Phi(j)$. The bound is attained exactly when every position takes exit j .

Proof. the upsample and the head run once per frame regardless of the map, every position runs at least $(j+1)\beta$ blocks because the feasible set starts at j , and every position that is not at the deepest exit wears an adapter. Summing and using $C_{\text{true}} \geq C_{\text{sep}}$ gives the bound; on a constant field the two costs agree, which gives attainment. ■

Table 1. Proposition 8 instantiated with the measured constants $u=0.0816$, $p=0.07453$, $\beta=2$, $a=0.04645$, $h=0.024$, $\sigma=0.0095$. The split depth costs 29.8 points of ceiling, and seam repair a further 0.95.

configuration	Φ	ceiling
tiled, $j=2$ (with seam repair)	0.6087	39.13%
per position, $j=2$	0.5992	40.08%
per position, $j=1$	0.4502	54.98%
per position, $j=0$	0.3011	69.89%

Proposition 9 (feasibility of the ceiling). *At a distortion budget β_{dB} the ceiling is attainable if and only if the uniform allocation $d \equiv j$ already meets the budget. Otherwise the optimum lies at an interior λ and the reachable saving is strictly below the ceiling.*

Proof. the ceiling is attained only by $d \equiv j$ (Proposition 8), so it is feasible exactly when that allocation is. If it is not, the constraint binds, and by Proposition 5 the constrained optimum is the λ at which the budget holds with equality. ■

On the measured system the uniform- j allocation costs 0.107 dB at the lowest rate and 0.248 at the highest, against a 0.1 dB budget, so Proposition 9 says the ceiling is not reachable at that budget at any rate — which is why the realised saving sits at 70–90% of it and why a looser 0.3 dB budget saturates instead of improving.

8 What the three changes are, in these terms

The companion report measures three inference-time changes. In the language above they are exactly:

- (i) Replace the tiled decode by the dilated one. This converts Assumption 2 of the main paper into Theorem 2, removes σ from Φ by Corollary 2, and replaces the additive cost by C_{true} , which exceeds it by the band.
- (ii) Refine the depth field. Under tiling the cell size is bounded below by the seam it creates; under Corollary 2 there is no such bound, and by Proposition 3 the price of refinement is the perimeter term alone. The optimum over cell sizes is interior and was located empirically at 64 px.
- (iii) Take $j = 0$. The split exists to bound the seam; by Corollary 2 the seam is identically zero, so the constraint has no purpose and Proposition 8 gives back 29.8 points of ceiling. Whether that translates into realised saving depends on whether the shallow rungs are good enough for Proposition 9 to bite less hard, which is an empirical question the companion report answers per rate.

9 Relation to the main paper's theory

The main paper proves that per-tile assignment achieves the Minkowski average of the per-tile achievable sets, that the adaptivity gain $\Delta(\lambda)$ is non-negative and vanishes only when one exit is optimal for every tile, and that the frontier is convex in the units it is plotted in. All three survive verbatim with tiles replaced by positions, because they depend only on separability of the objective — which Theorem 2 supplies exactly where the tiled version assumed it. What does not survive is cost additivity, and Theorem 4 is the replacement: the true cost is convex, the additive form is its tight linear minorant, and the allocator that uses the minorant is suboptimal by at most λ times a band it can measure.

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